

TIM HORTONS

Business Performance Dashboard

Business Requirements Document (BRD)

A structured business requirements specification defining the purpose, stakeholders, analytical questions, KPI requirements, functional requirements, non-functional requirements, dashboard scope, acceptance criteria and future roadmap for the Power BI solution.

Metric	Requirement Baseline	Metric	Requirement Baseline
Revenue	\$254,579.62	Profit	\$95,611.88
Orders	17,444	Units Sold	43,529
Profit Margin	37.6%	Average Order Value	\$14.59
Stores	6	Cities	5
Categories	5	Products	19

Document status: Portfolio / analytical project specification **Reporting platform:** Microsoft Power BI **Data period:** 2024

Purpose of this BRD: translate the business need into clear, testable reporting requirements so that the dashboard can be evaluated as a business solution rather than only as a visual design.

1. Business Context

The business requires a centralized analytical view of sales and operating performance across products, categories, customers, stores, geography and time. The existing transactional dataset contains detailed sales information but requires transformation into decision-ready management information.

The proposed Power BI dashboard addresses this need through a four-page analytical experience: **Executive Overview**, **Product & Category Analysis**, **Customer & Sales Behavior**, and **Store & Geographic Analysis**.

1.1 Business Problem Statement

- Management needs a single view of revenue, profit, order volume and units sold.
- Decision-makers need to understand which categories/products generate the most value.
- Customer-type and transaction-timing patterns need to be visible for commercial planning.
- Store and geographic concentration needs to be identified for benchmarking and growth decisions.
- Users need interactive filters to move from high-level performance to specific business segments.

Problem statement: Without an integrated analytical dashboard, transactional data is difficult to interpret consistently across financial, customer, product and geographic dimensions.

2. Business Objectives

Objective ID	Business Objective	Success Indicator
BO-01	Provide a single executive view of business performance.	Core KPIs available on one page.
BO-02	Monitor revenue, profit and profitability trends.	Monthly revenue/profit trend and margin KPI.
BO-03	Identify high-value products and categories.	Top-5 rankings and category comparison.
BO-04	Understand customer behavior and mix.	New vs returning analysis, timing and payment views.
BO-05	Evaluate store and geographic performance.	Region, city, store and map analysis.
BO-06	Enable self-service filtering.	Slicers for date/month, region, store, category and customer type.
BO-07	Support action-oriented decision-making.	Insights and recommendations traceable to visuals.

Primary business outcome: improve the speed and consistency with which stakeholders can identify performance drivers, exceptions and opportunities.

3. Stakeholders & User Roles

Stakeholder / Role	Information Need	Dashboard Usage
Executive / Management	Revenue, profit, margin, orders and major trends.	Executive Overview.
Sales / Commercial Team	Product, category, customer and timing patterns.	Product & Customer pages.
Store / Operations Team	Store, city, region and demand patterns.	Store & Geographic page.
Marketing / Customer Team	New vs returning behavior and sales timing.	Customer & Sales Behavior.
BI / Data Analyst	Model integrity, KPI definitions and drill/filter behavior.	All pages + data model.
Portfolio Reviewer / Recruiter	Evidence of analytical workflow and business thinking.	Overall solution and documentation.

4. Project Scope

4.1 In Scope

- Historical sales analysis for the supplied 2024 transactional dataset.

- Revenue, profit, orders, units, margin and average order value.
- Product and category performance analysis.
- New vs returning customer-type analysis.
- Day-of-week and time-of-day sales behavior.
- Store, city, province/region and geographic analysis.
- Interactive slicers and page navigation.
- Power BI star-schema modeling and reusable DAX measures.

4.2 Out of Scope

- Real-time transaction streaming.
- Inventory availability and stock forecasting.
- Employee/staffing performance.
- Marketing campaign attribution.
- Customer-level churn or lifetime value without a persistent CustomerID.
- Financial budgeting/forecasting unless target data is separately supplied.

5. Business Questions the Dashboard Must Answer

Question ID	Business Question	Primary View
BQ-01	What are total revenue, profit, orders, units and profit margin?	Executive Overview
BQ-02	How are revenue and profit changing over time?	Executive Overview
BQ-03	Which stores generate the most revenue?	Executive / Store
BQ-04	Which categories and products drive revenue and profit?	Product & Category
BQ-05	What is the revenue mix across categories?	Product & Category
BQ-06	How does new vs returning customer-type revenue compare?	Customer & Sales
BQ-07	When during the week and day does demand occur?	Customer & Sales
BQ-08	Which payment methods contribute most to profit/revenue?	Customer & Sales
BQ-09	Which regions/cities have the greatest revenue concentration?	Store & Geographic
BQ-10	Which stores should be benchmarked for best practice?	Store & Geographic

6. Functional Requirements

6.1 Executive Overview Requirements

Req ID	Requirement	Priority	Acceptance Test
FR-EO-01	Display Total Revenue, Total Profit, Total Orders, Units Sold and Profit Margin.	Must	All five KPIs render correctly and respond to filters.
FR-EO-02	Display monthly Revenue and Profit trend.	Must	Trend changes when date filters are applied.
FR-EO-03	Rank stores by revenue.	Must	Stores sort descending by revenue.
FR-EO-04	Rank categories by revenue.	Must	All five categories can be compared.
FR-EO-05	Display top five products by revenue.	Should	Exactly five products shown under default context.

6.2 Product & Category Requirements

Req ID	Requirement	Priority	Acceptance Test
FR-PC-01	Show Top 5 Products by Revenue.	Must	Ranking updates with slicers.
FR-PC-02	Show Top 5 Products by Profit.	Must	Ranking uses profit measure, not revenue.
FR-PC-03	Show Category Revenue Mix.	Must	Category contribution is visible and filter-responsive.
FR-PC-04	Show category performance by revenue, profit and units.	Must	All categories are comparable.
FR-PC-05	Provide category summary table with ASP and margin.	Should	Table totals reconcile to headline KPIs.

6.3 Customer & Sales Behavior Requirements

Req ID	Requirement	Priority	Acceptance Test
FR-CS-01	Display total, returning and new customer-type transaction counts.	Must	Counts respond to filters.
FR-CS-02	Compare revenue by customer type.	Must	New vs returning revenue is visible.
FR-CS-03	Show revenue by day of week.	Must	Mon-Sun comparison is available.
FR-CS-04	Show revenue by time of day/hour.	Must	Time periods are chronologically sorted.
FR-CS-05	Show payment-method performance.	Should	Payment methods are ranked by value.
FR-CS-06	Show monthly customer-type revenue trend/percentage.	Should	Monthly series responds to filters.

6.4 Store & Geographic Requirements

Req ID	Requirement	Priority	Acceptance Test
FR-SG-01	Show store count and top-performing store.	Must	Top store updates with filter context.
FR-SG-02	Show revenue by region.	Must	Regional comparison is visible.
FR-SG-03	Show top five stores by revenue.	Must	Ranking updates dynamically.
FR-SG-04	Show revenue by province/city geographically.	Should	Map reflects selected context.
FR-SG-05	Show top cities by revenue and city heatmap.	Should	City comparison is visible.

7. Filter, Navigation & Interaction Requirements

Req ID	Requirement	Priority
UX-01	Provide consistent page navigation across all analytical pages.	Must
UX-02	Provide date/month filtering where relevant.	Must
UX-03	Provide Store, Category and Customer Type slicers.	Must
UX-04	Provide Region slicer on relevant pages.	Should
UX-05	All visuals should respond consistently to slicer selections unless intentionally excluded.	Must
UX-06	Use clear visual titles and business-friendly KPI names.	Must
UX-07	Maintain consistent Tim Hortons-inspired branding and accessibility-conscious contrast.	Should
UX-08	Avoid unnecessary visual clutter; prioritize decision-useful charts.	Should

8. Data Requirements

Requirement	Specification
Source	Supplied Tim_Hortons_Portfolio_Dataset(1).xlsx
Fact grain	One transactional record/order row.
Time coverage	01-Jan-2024 through 30-Dec-2024.
Core dimensions	Date, Product, Category, Store, City/Location, Customer Type, Payment Method.
Core measures	Revenue, Cost, Profit, Quantity, Discount, Unit Price.
Data model	Star-schema approach with SalesData as fact and dimensions for descriptive analysis.
Data refresh	Manual/file-based refresh for portfolio implementation; no real-time requirement.
Data quality	Numeric, date/time and categorical fields must have valid types and aggregation behavior.

8.1 Data Governance Requirement

KPI definitions must remain consistent across pages. The same Revenue, Profit, Orders, Units Sold, Margin and AOV measures should be reused rather than recreated independently in each visual.

Customer analytics caveat: CustomerType is available, but a persistent CustomerID is not. Therefore the solution must describe customer KPIs as customer-type/transaction behavior unless a true customer identifier is introduced.

9. Non-Functional Requirements

Category	Requirement	Acceptance Criteria
Usability	Dashboard should be understandable by a business user without technical knowledge.	KPI labels, titles and navigation are self-explanatory.
Performance	Interactions should feel responsive for the supplied dataset.	Filters and visuals update without obvious unnecessary delay.
Consistency	Measures and formatting should remain consistent across pages.	Same KPI produces same result everywhere.
Visual Design	Use a coherent visual hierarchy and restrained palette.	Red/cream branding, aligned cards and consistent typography.
Maintainability	Model and measures should be reusable and organized.	Dimensions separated from fact data; measures documented.
Scalability	Design should support future dates and additional transactions.	Date dimension and model structure can accept additional data.
Accuracy	Dashboard totals must reconcile to source data.	Headline values match validated source calculations.
Security	No sensitive personal customer information is required for current scope.	Dataset contains no requirement for PII.

10. KPI Requirements & Targets

The following KPIs are required for the initial release. Targets are not included because no budget/target dataset was supplied.

KPI	Required	Current Baseline	Decision Use
Total Revenue	Yes	\$254,579.62	Top-line scale
Total Profit	Yes	\$95,611.88	Absolute profitability
Profit Margin	Yes	37.6%	Profit efficiency
Total Orders	Yes	17,444	Transaction volume
Units Sold	Yes	43,529	Demand/product volume
Average Order Value	Yes	\$14.59	Basket value
Average Price	Yes	\$5.85	Pricing/realized value
Store Revenue	Yes	Dynamic	Location performance
Category Revenue	Yes	Dynamic	Portfolio mix

11. Overall Acceptance Criteria

- All five executive KPI cards calculate correctly against the source data.
- Revenue and profit trend visuals reconcile to the underlying transaction totals.
- Product/category rankings correctly sort according to the selected measure.
- Customer-type visuals distinguish New and Returning records correctly.
- Day-of-week and time-of-day visuals follow logical chronological ordering.
- Store, region, city and category filters affect relevant visuals consistently.
- The four report pages are reachable through visible navigation.
- No visual contains misleading duplicate aggregation or an unexplained blank category.
- Totals in detailed tables reconcile with headline KPIs under the same filter context.
- The report remains understandable when opened with default 'All' filter selections.

12. Risks, Assumptions & Dependencies

Type	Item	Mitigation / Treatment
Risk	No persistent CustomerID.	Treat CustomerType analysis as a proxy; add CustomerID in future.
Risk	No targets/budgets supplied.	Use descriptive performance analysis; add targets for variance reporting.
Risk	Geographic data may be limited to city/province/region.	Standardize geographic keys and coordinates for production maps.
Assumption	Revenue and Profit fields are authoritative for this portfolio dataset.	Validate against source totals before each release.
Dependency	Power BI desktop/model relationships and DAX measures.	Document model and measures; test interactions after changes.
Dependency	Source Excel workbook structure remains stable.	Use controlled column names and transformation steps.

13. Future Requirements / Enhancement Roadmap

Phase	Enhancement	Business Value
Phase 1	CustomerID and customer-level purchase history.	True retention, frequency, churn and CLV.
Phase 2	Sales targets/budgets.	Actual vs target and variance analysis.
Phase 3	Inventory and staffing data.	Operational root-cause analysis.
Phase 4	Promotion/campaign data.	Campaign ROI and attribution.
Phase 5	Multi-year history.	YoY growth, seasonality and forecasting.
Phase 6	Automated refresh / Power BI Service deployment.	Operational self-service reporting.

14. Requirements Traceability Matrix

This matrix links the business objective to the dashboard requirement and resulting analytical output.

Business Objective	Requirement Group	Dashboard Output	Status
BO-01 Executive visibility	FR-EO-01 to FR-EO-05	Executive Overview	Defined
BO-02 Financial trends	FR-EO-02	Revenue/Profit trend	Defined
BO-03 Product optimization	FR-PC-01 to FR-PC-05	Product & Category	Defined
BO-04 Customer behavior	FR-CS-01 to FR-CS-06	Customer & Sales	Defined
BO-05 Geographic performance	FR-SG-01 to FR-SG-05	Store & Geographic	Defined
BO-06 Self-service filtering	UX-01 to UX-06	Slicers/navigation	Defined
BO-07 Actionable decisions	All requirements	Insights/recommendations	Defined

15. Sign-Off Checklist

Review Area	Complete When...
Business scope	All in-scope questions are represented in the dashboard.
Data	Source totals and data types are validated.
KPIs	Definitions are documented and reconciled.
Visuals	Each visual has a clear business purpose.
Interactions	Slicers and navigation behave consistently.
Acceptance	All mandatory acceptance criteria pass.
Documentation	BRD and technical documentation are stored with the project.

Final requirement statement: The completed Power BI solution must provide a reliable, interactive and business-oriented view of Tim Hortons sales performance across financial, product, customer, temporal and geographic dimensions, while remaining maintainable and extensible for future data.

Portfolio note: This BRD documents the supplied analytical portfolio project and is not an official corporate Tim Hortons requirements document.